# IS883: Deploying Generative AI

Mohannad Elhamod

BOSTON UNIVERSITY

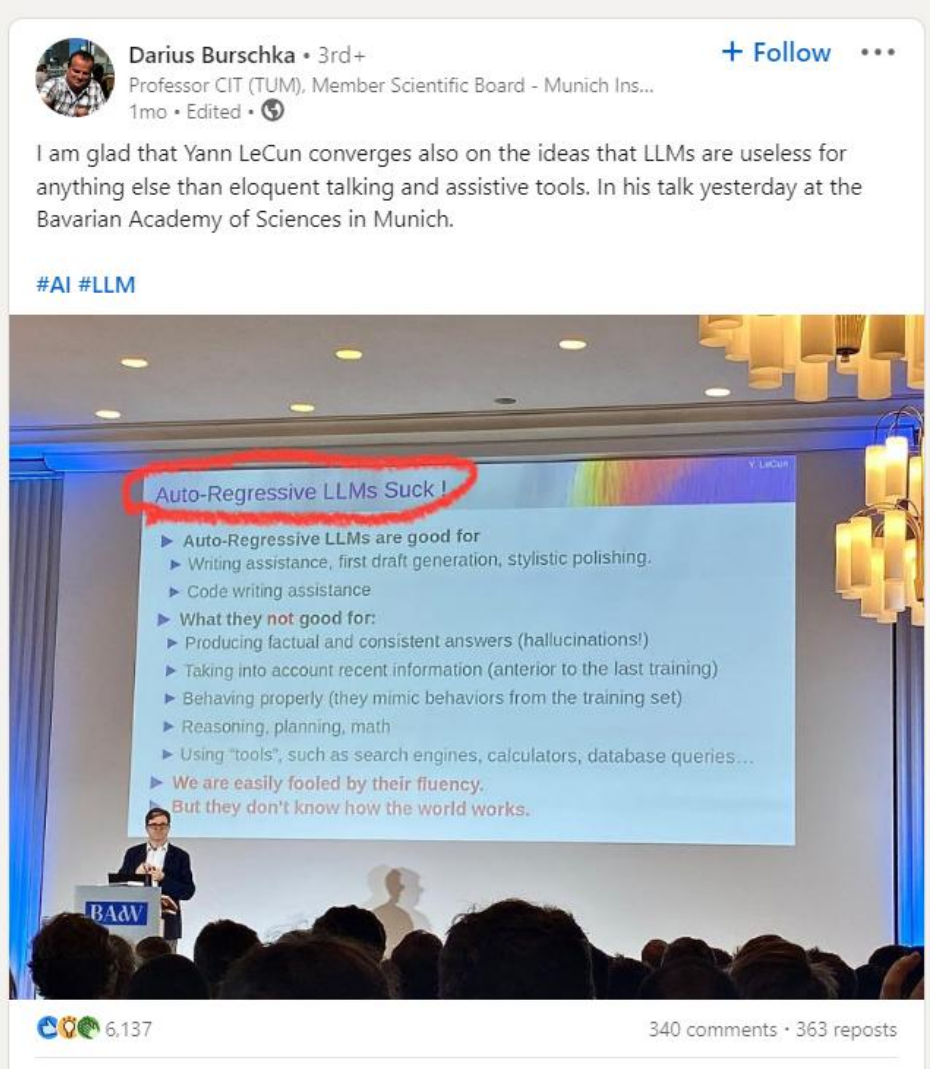
**Boston University** Questrom School of Business

# Hallucinations

BOSTON UNIVERSITY

# Why Do LLMs Hallucinate?

- Language models are just glorified text completion tools.
- They are highly influenced by:
  - Data quality
  - Input context.



# Examples

- Walk or take the car to the car wash?
- Fake travel guides




NYTimes

# Minimizing Hallucinations

- Parameter tuning (e.g., temperature).
- Prompt engineering.
- Reasoning Frameworks.

# Reasoning

BOSTON UNIVERSITY

# Example

User • 12:16 AM

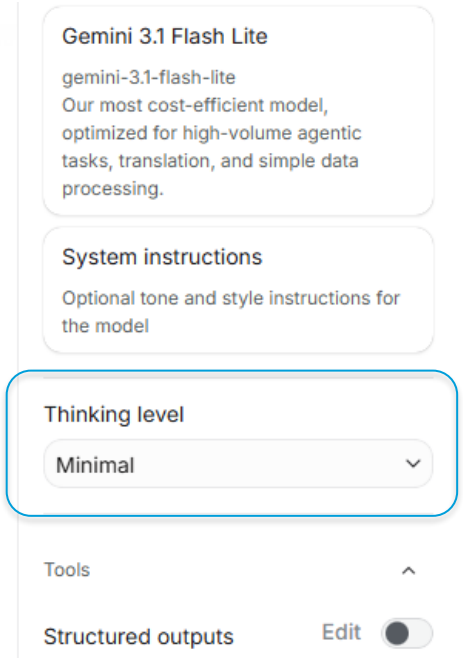
If block C is on top of block A, and block B is separately on the table, can you tell me how I can make a stack of blocks with block A on top of block B and block B on top of block C, but without moving block C?

Model • 12:16 AM

To achieve this configuration without moving block C, you must work around its current position. Since block C is already at the bottom of the stack you desire (C is the base for B, which is the base for A), you can accomplish this by moving the other blocks as follows:

1. **Remove block A** from the top of block C and place it aside temporarily.
2. **Move block B** from the table and place it directly on top of block C.
3. **Place block A** on top of block B.

**Result:** You now have a stack where A is on top of B, and B is on top of C, all without ever having moved block C.



# Reasoning to The Rescue!

User • 12:19 AM

If block C is on top of block A, and block B is separately on the table, can you tell me how I can make a stack of blocks with block A on top of block B and block B on top of block C, but without moving block C?

Model • 12:19 AM

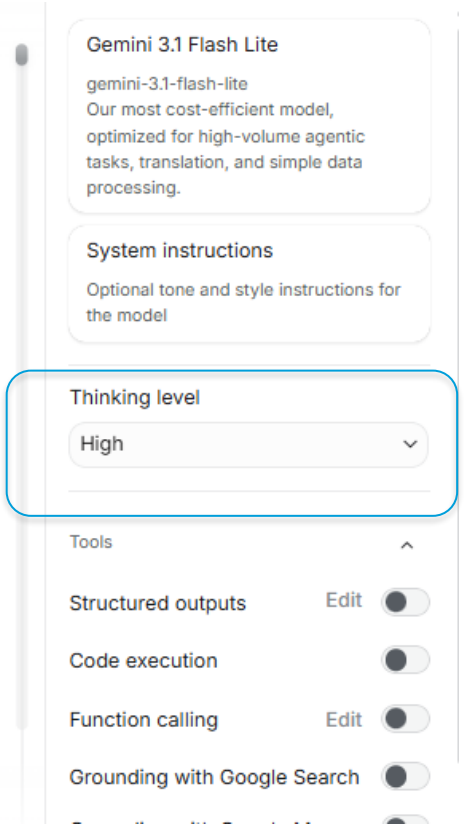
Thoughts

Expand to view model thoughts

It is **physically impossible** to create that stack without moving block C.

Here is why:

1. **The current position of C:** Currently, block C is sitting on top of block A.
2. **The goal position of C:** To have a stack where block B is on top of block C (making C the bottom-most block), you would have to move block C from its current position on top of block A and place it on the table.
3. **The blockage of A:** Because block A is currently underneath block C, you cannot get block A out to place it at the top of the stack without moving block C out of the way first.



# Prompt Engineering

BOSTON UNIVERSITY

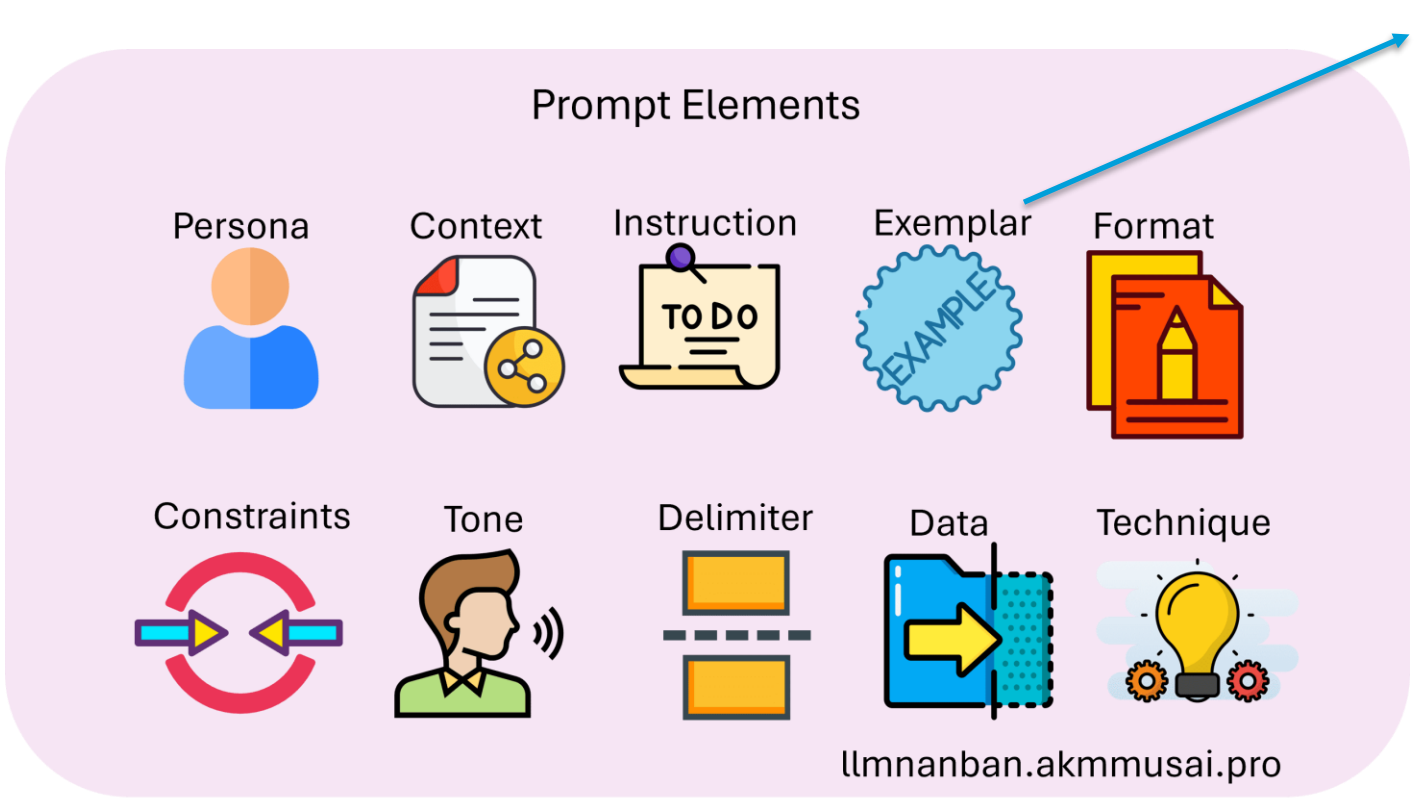
**Boston University** Questrom School of Business

# Why Need Prompt Engineering?

Let's say you have a pre-trained model to be used for an airline chat agent use-case :

- If the model has not already been trained on some similar chat agent data, one approach would be to fine-tune it.
- What if your model has already been trained on such data but you are not getting the desired output?
  - you probably need to better *engineer* your prompt to extract the desired knowledge/pattern.

# Elements of Prompt Engineering




**The more specific you are, the better the results!**

# Example

```
<persona>
[Who AI should act as]
</persona>

<context>
[Background information about your situation]
</context>

<tone>
[How it should sound]
</tone>

<input-format>
[What kind of information you're providing]
</input-format>

<examples>
[Sample of what good output looks like]
</examples>

<task>
[Exactly what you want it to do]
</task>

<constraints>
[Limitations or requirements]
</constraints>

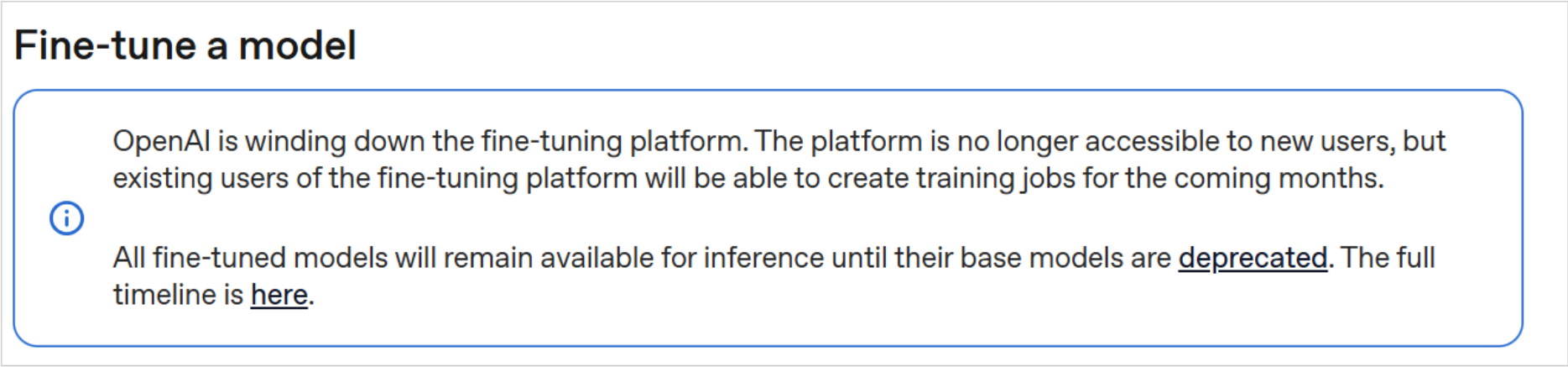
<response-format>
[How you want the output structured]
</response-format>
```

# Prompt Engineering vs Fine-Tuning

| | Prompt engineering | Fine-tuning |
|---|---|---|
| Technical Expertise Needed | Low | High |
| Cost | Low upfront/ High long-term | High upfront/ Low long-term |
| Effectiveness | Moderate | High |

# Can You Still Fine-Tune?

- Aug 2026: OpenAI is winding down its hosted fine-tuning platform — closed to new users; existing users get “the coming months.”
- This is one vendor’s product decision, not a verdict on the method: open-weight fine-tuning (Hugging Face, LoRA) and other providers still offer it.




Source: developers.openai.com/api/docs/guides/model-optimization — retrieved 22 Aug 2026

# Pushing the Idea Further: Projects in ChatGPT

- Let's try it!